

Validating Billow against a hanging V3 kite

Measured deformed shape, ten load cases, and what the dataset can and cannot decide

Structural module, AWETrim

September 11, 2026

Abstract

The TU Delft V3 kite was suspended from its bridle and its deformed shape measured under ten combinations of inflation pressure and applied load. This report rebuilds that experiment in **Billow**, the standalone minimum-energy structural model in `awetrim.structural`, and scores it against both the measurements and the reference finite-element model `kite_fem`, on an element-for-element identical mesh.

Three things came out of it. First, the distributed reference model is *ill posed*: its anchor node belongs to no element, so the kite is a free body under 126 N of unbalanced weight, and the reference results are held together by a numerical regularisation. Second, `kite_fem`'s beams *converge* to about $0.23\times$ its own constitutive law, with negative bending stiffness in several load cases — so its closer agreement with the measurement is not evidence of better physics. Third, and most limiting for the original purpose, this dataset is governed almost entirely by inflatable-tube bending and is nearly blind to the canopy model, so no canopy conclusion can be drawn from it.

Billow reaches force balance to $\sim 1 \times 10^{-9}$ N in a mean of 14 seconds per case against the reference's 552 s. On the six cases both codes can solve, the mean relative error is 11.5% for Billow against 7.8% for `kite_fem`; over all ten it is 15.7% against 12.4%, the difference being four centre-load cases that neither code reproduces — the thesis says the same of its own model. Neither reproduces the measured shape — though the published data, six independent numbers with no height or chordwise measure among them, is a weak test of shape at all. The report closes with the sharpest remaining clue: the measured pressure sensitivity is roughly five times what the fitted tube law predicts, at pressures below the range over which that fit was ever demonstrated.

1 The experiment

The kite hangs from its bridle and the deformed shape is recorded stereoscopically, with markers on the inflatable structure. Twelve lengths are extracted per load case: nine trailing-edge segments `La-Li` between consecutive struts, the tip-to-tip span `b`, and two tip-to-centre-leading-edge diagonals `LcsTL`, `LcsTR`.

The deformation is large. Under self-weight alone the span falls from 8.305 m as built to a measured 4.867 m — a 41% closure — and every trailing-edge segment shortens by 26–43%. The tip-load cases run the same deformation backwards: spreading the tips with 2 kg or 5 kg reopens the span to 7.8–8.3 m, close to the as-built shape.

Suspension. The kite hangs from a **bar**, not a point. Roeleveld's thesis (Appendix B, Table B.1) marks `A_main`, `A_I`, `M`, `PL` and `SL` as N/A for the hanging test and adds a **bar** of 1650 mm; Section 6.3 explains that the bridle was adapted to fit the hangar height. The same table records the one documented shortening, `B_r,5` from 12 168 to 9425 mm.

Provenance caveat. The upstream repository describes the measurements only as “experimental validation reference data” and never uses the word photogrammetry; the thesis identifies the method

as stereoscopic camera measurement. The published rows are also perfectly symmetric left-to-right to four decimals, so they have been mirrored or port/starboard-averaged.

2 Rebuilding the model

Billow is rebuilt from `hanging_test_initial.npz`, the fully assembled reference model, rather than from the YAML — there is no committed YAML→npz builder, so the `.npz` is the input both codes solve. Nodes, connectivity, rest lengths, stiffnesses and tube diameters therefore transfer with no re-interpretation:

shared, element for element	276 nodes, 46 bridle cables, 14 pulleys, 98 inflatable tube beams
changed	canopy: 721 tension-only springs → 392 wrinkling membrane triangles

Two checks make this trustworthy rather than assumed. The load vector reproduces `kite_fem`'s own `loading()` to **exactly zero** for all five load combinations, and the length extractor reproduces its published `model_results.csv` from its saved states to **machine zero**.

Canopy grid. The reference canopy is a structured 29×8 grid stored only as a flat spring list. It is recovered geometrically — the reference builder projects every interior section node onto the straight line joining its leading- and trailing-edge node, so each chordwise section is exactly collinear — and then *verified*: all 29 sections must come out with 8 nodes, and all 721 canopy springs must be grid edges with no spring joining two grid nodes being anything else. Both hold exactly.

3 Defects found in the reference model

3.1 The anchor is connected to nothing

Node 0 is flagged fixed but appears in no spring, no pulley and no beam; seven further bridle nodes are orphaned with it (all of exactly zero mass). The kite is therefore a free body under 126 N of unbalanced weight, and its potential is unbounded below. Billow reports that faithfully: the structure translates hundreds of kilometres, and neither load stepping nor constitutive continuation helps, because none of them supplies the missing constraint.

`kite_fem` does not diverge only because its `I_stiffness` regularisation acts as a ground spring on every node. Its own published solutions still drift 0.29–0.34 m bodily upward, with node 0 sitting fixed and unconnected throughout.

3.2 Bridle lengths disagree with the thesis

Twenty-eight of the thirty-six declared bridle lines match Table B.1 to the millimetre, which is what makes the remaining eight look like transcription errors rather than naming mismatches: `BRI` and `BRII` are an exact swap; `abcd2` and `abcd3` carry a duplicated leading digit; `ab3` equals `ab2`; `a5` and `br5` differ; and `brmain_ext` is present in the table but unwired.

The eighth is not a transcription error but a contradiction inside the thesis itself. Table B.1 marks `A.I` as N/A for the hanging test, while Figure B.1 *draws* `A.I` running to the bar in the adapted geometry. The two readings are structurally different — cutting the line removes a front load path — so both were run:

3.3 The reference beams converge far softer than their own law

Billow's integrated tube law agrees with `kite_fem`'s secant EI to 1–3% across the whole curvature range, departing only below its 0.03 normalised-deflection floor. The two codes therefore implement the same constitutive law. They do not converge to the same stiffness:

Table 1: Three readings of the same source. Cutting A_I is the most accurate reading on the cases the model can solve and the worst overall, because removing that load path is what lets the four centre-load cases fall into a folded local minimum. No reading is adopted here: the ambiguity is a finding about the source data, not something to settle by taking whichever scores best.

bridle reading	all ten (%)	six solvable (%)	four centre-load (%)	folded cases
bridle as stored	15.1	11.0	21.3	0
Table B.1, A_I cut	17.3	8.9	30.0	4
Table B.1, A_I kept	15.7	11.5	21.9	0
<code>kite_fem</code>	12.4	7.8	19.3	0

load case	median EI (Nm ²)	vs. own law	min EI	beams flagged collapsed
1	24.5	0.29×	2.7	78
4	69.4	0.81×	−42.4	49
6	43.1	0.50×	−164	76
9	84.1	0.98×	−459	40

Three of the four cases contain *negative* bending stiffness, which the secant form $EI = PL^3/(3(v - PL/kGA))$ produces as soon as the deflection estimate falls below the shear term. `kite_fem` therefore lands closer to the measurement than Billow because its beams are several times softer than the law it implements, not because its physics is closer.

4 Results

Table 2: All ten load cases. Errors are the mean over the twelve measured lengths of $|L_{\text{model}} - L_{\text{meas}}|/L_{\text{meas}}$; the reference `shapecorrelation` metric is not used, because as a dot-product ratio dominated by the two quantities of order 4–8 m it reads 0.980–0.9997 even where the span is 48% wrong. $\|r\|$ is the force residual on the worst free node.

LC	load case			span (m)			mean rel. error (%)		Billow	
	p (bar)	tip (kg)	centre (kg)	meas.	Billow	<code>kite_fem</code>	Billow	<code>kite_fem</code>	t (s)	$\ r\ $ (N)
1	0.15	0	0	4.867	7.276	5.382	25.9	13.7	7	2.2e-07
2 [†]	0.15	0	9.7	5.042	7.554	7.462	27.6	25.0	15	9.1e-08
3 [†]	0.15	0	25.2	5.113	7.841	7.576	21.8	18.9	19	3.8e-07
4	0.15	2	0	7.843	8.597	8.467	6.5	6.7	16	1.1e-07
5	0.15	5	0	8.271	8.916	8.962	5.9	5.4	18	4.1e-07
6	0.25	0	0	5.453	7.468	6.363	19.5	9.3	6	4.0e-09
7 [†]	0.25	0	9.7	5.575	7.690	7.691	19.7	18.9	15	3.7e-09
8 [†]	0.25	0	25.2	5.697	7.965	7.638	18.4	14.6	17	6.4e-09
9	0.25	2	0	7.818	8.593	8.461	5.2	5.6	15	1.1e-07
10	0.25	5	0	8.224	8.908	8.955	6.2	5.9	16	4.0e-07
mean, all ten							15.7	12.4	14	
mean, excluding centre-load cases							11.5	7.8	13	

4.1 The solved kite

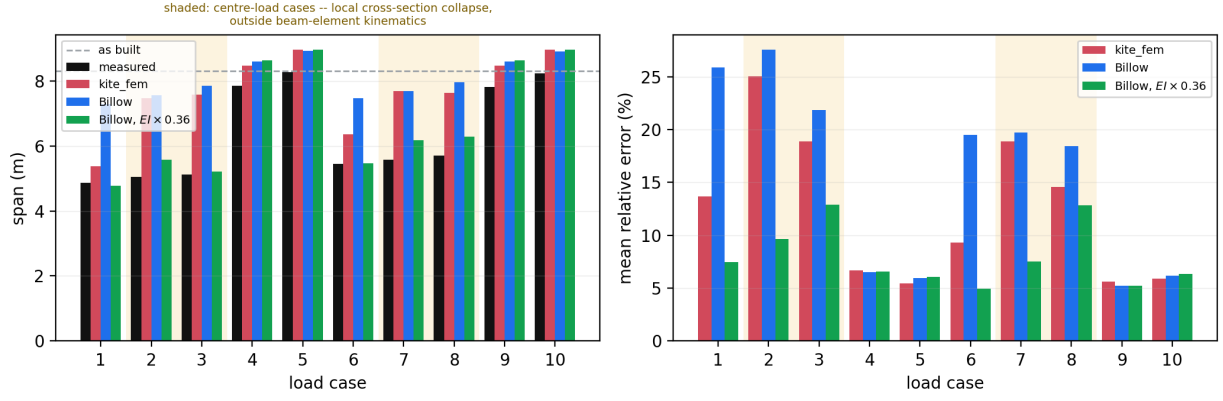


Figure 1: Span and mean relative error across the ten cases. Both codes overstate the span wherever they solve the physical branch; the gap is widest where gravity alone sets the shape and narrowest where an imposed tip load dominates. The shaded centre-load cases are reproduced by neither code, and are where Billow’s folded branch shows as a near-zero span.

Table 3: Per-quantity error, averaged over the ten cases. The two codes fail differently, which the signed-bias columns separate from the magnitude columns: Billow’s bias is one-signed — it overstates essentially every length, i.e. its wing stays too open — whereas `kite_fem` scatters, understating the outboard segments and the tip-to-centre diagonals while overstating the inboard ones. A single stiffness factor could plausibly account for the first and could not account for the second.

quantity	meas. range (m)	rel. error (%)		signed bias (%)	
		Billow	kite_fem	Billow	kite_fem
La	0.848–0.903	2.4	2.4	+1.9	-0.6
Lb	0.979–1.280	16.5	10.6	+16.5	+8.3
Lc	0.903–1.276	18.8	14.3	+18.8	+9.7
Ld	0.749–1.259	24.5	20.8	+24.5	+20.8
Le	0.920–1.217	13.0	16.2	+13.0	+16.2
Lf	0.749–1.259	24.1	20.8	+24.1	+20.8
Lg	0.903–1.276	18.4	14.3	+18.4	+9.7
Lh	0.979–1.280	16.9	10.6	+16.9	+8.3
Li	0.848–0.903	2.4	2.4	+1.9	-0.6
b	4.867–8.271	30.3	22.9	+30.3	+22.9
LcsTL	4.029–4.888	10.3	6.8	+10.3	+5.1
LcsTR	4.029–4.888	10.4	6.8	+10.4	+5.1

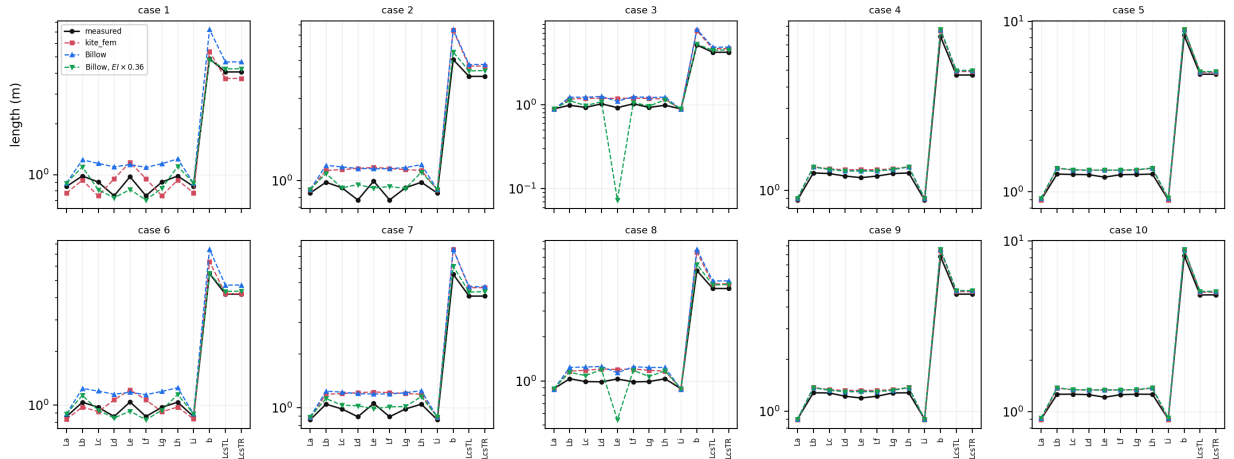
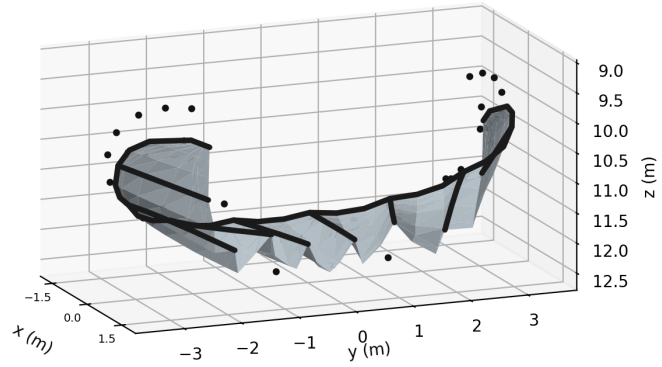


Figure 2: The twelve measured quantities per load case, model over measurement.

Load case 1: 0.15 bar | solved shape, measured markers overlaid

● measured markers



Load case 5: 0.15 bar, tip 5 kg | solved shape, measured markers overlaid

● measured markers

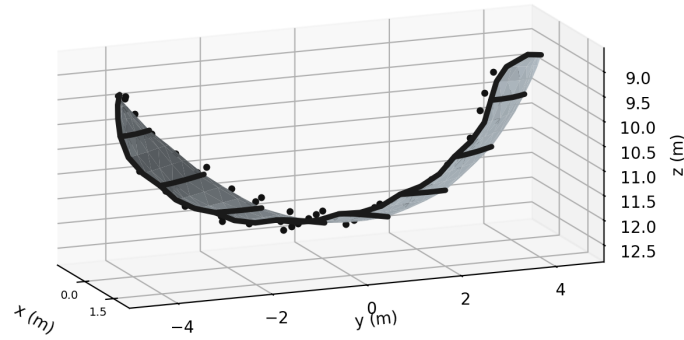


Figure 3: The solved shape under self-weight (top) and with the tips spread by 5 kg (bottom), with the measured markers overlaid. The canopy is shaded by facet orientation, so the billow between struts and the sag of each bay are visible directly. Both are the $EI \times 0.36$ solution.

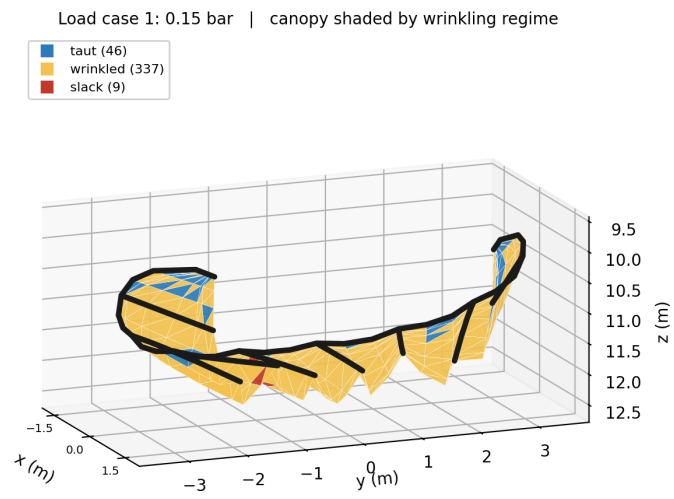


Figure 4: The same self-weight case with each canopy triangle coloured by its tension-field state. This is a prediction the membrane formulation makes and a tension-only spring net cannot: under self-weight almost the whole canopy is wrinkled — one principal stress relaxed to zero — with taut material only where the bridle pulls through, and a handful of fully slack triangles at the tip.

4.2 Geometry, case by case

The figures that follow show the inflatable structure — leading edge and the ten struts — in front, top and side projection, with the as-built shape for reference. The canopy is omitted for legibility; it carries little of the load in this case (Section 5).

Black dots are Haanen’s photogrammetry markers, carried into the model frame by a rigid transform. For these figures the registration uses the *leading edge alone*, so the two leading edges lie on top of one another and the remaining disagreement appears where it physically is — in the struts and the canopy — rather than being spread evenly over the structure by a global fit. The error table in Section 9 uses its own per-model alignment over all curves, which is the fair one for scoring; these overlays are for reading, not for measuring. The markers stop short of the tips because the outermost bays are not instrumented.

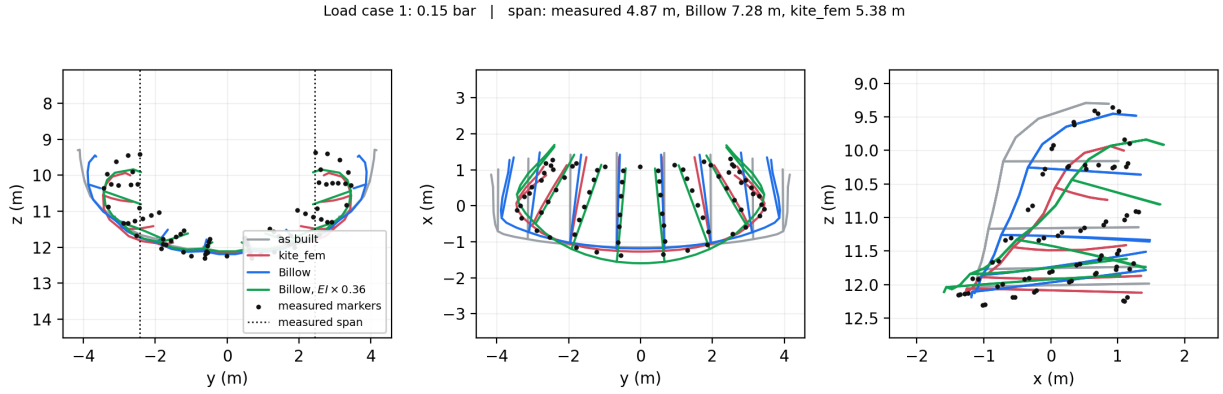


Figure 5: Load case 1.

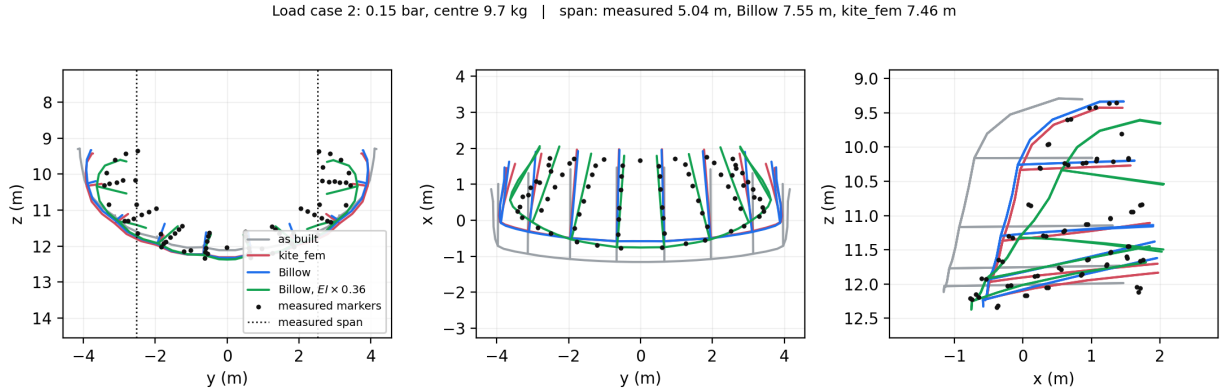


Figure 6: Load case 2.

Load case 3: 0.15 bar, centre 25.2 kg | span: measured 5.11 m, Billow 7.84 m, kite_fem 7.58 m

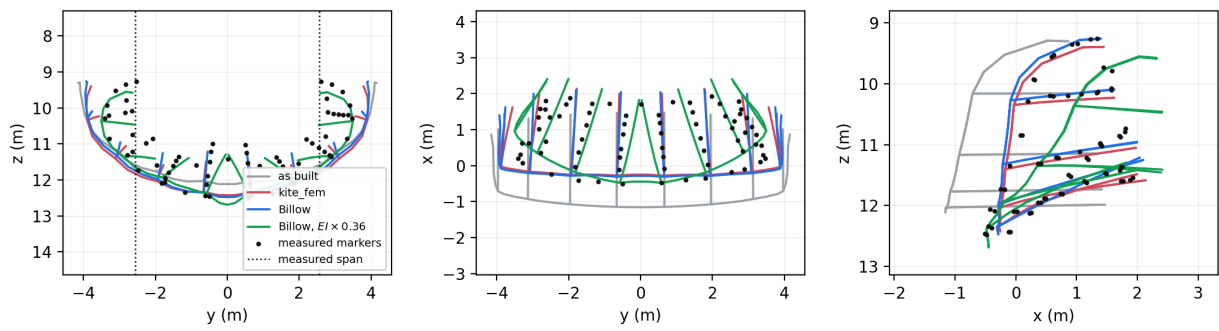


Figure 7: Load case 3.

Load case 4: 0.15 bar, tip 2 kg | span: measured 7.84 m, Billow 8.60 m, kite_fem 8.47 m

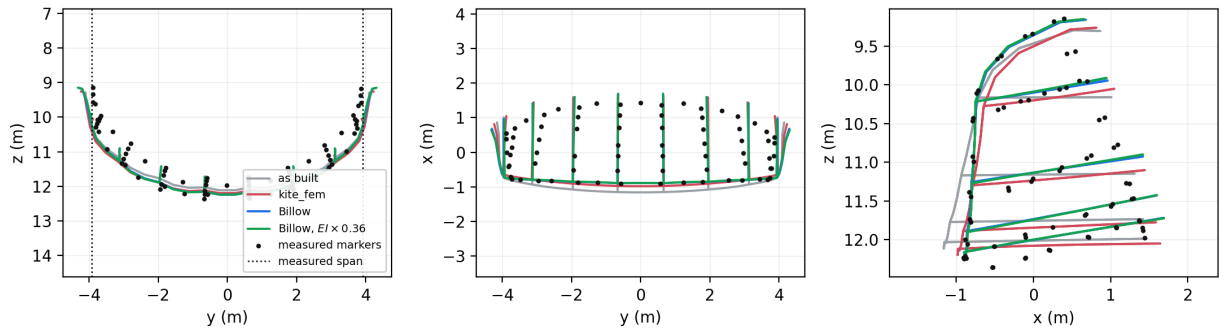


Figure 8: Load case 4.

Load case 5: 0.15 bar, tip 5 kg | span: measured 8.27 m, Billow 8.92 m, kite_fem 8.96 m

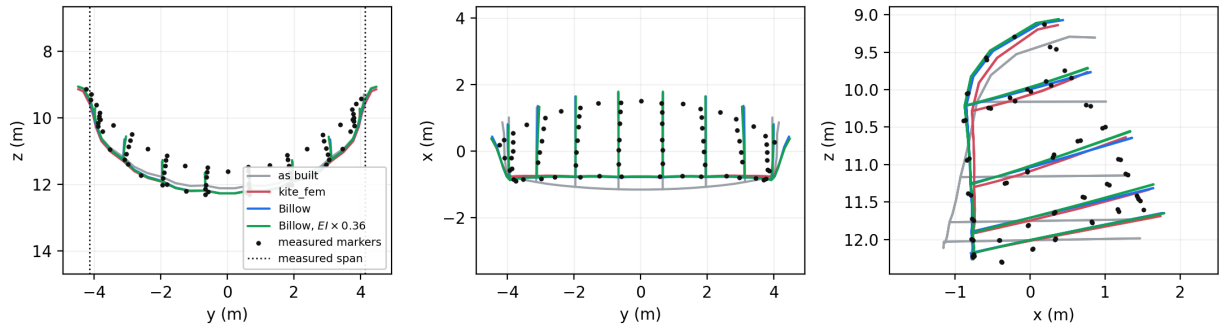


Figure 9: Load case 5.

Load case 6: 0.25 bar | span: measured 5.45 m, Billow 7.47 m, kite_fem 6.36 m

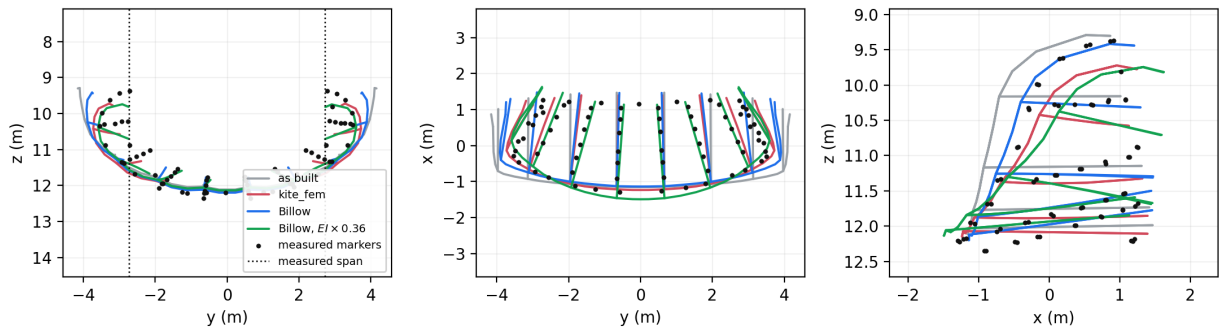


Figure 10: Load case 6.

Load case 7: 0.25 bar, centre 9.7 kg | span: measured 5.58 m, Billow 7.69 m, kite_fem 7.69 m

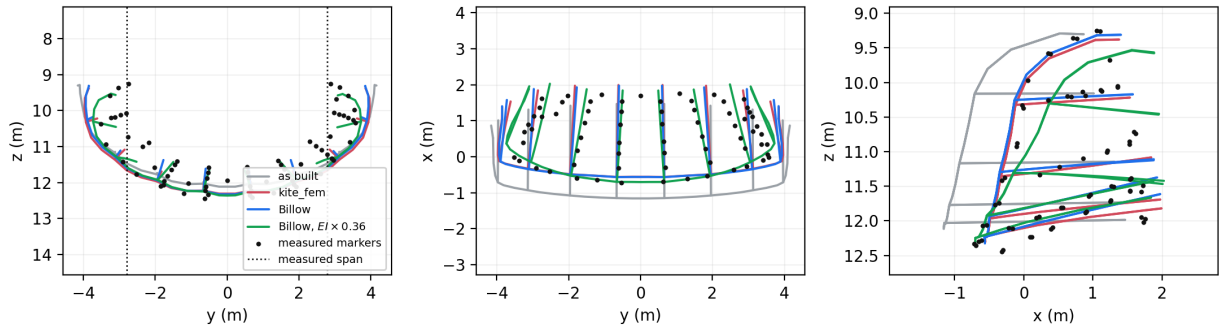


Figure 11: Load case 7.

Load case 8: 0.25 bar, centre 25.2 kg | span: measured 5.70 m, Billow 7.96 m, kite_fem 7.64 m

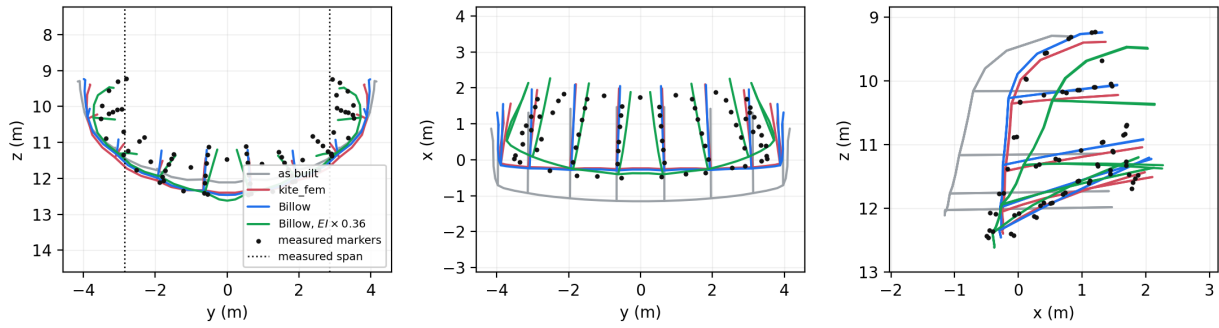


Figure 12: Load case 8.

Load case 9: 0.25 bar, tip 2 kg | span: measured 7.82 m, Billow 8.59 m, kite_fem 8.46 m

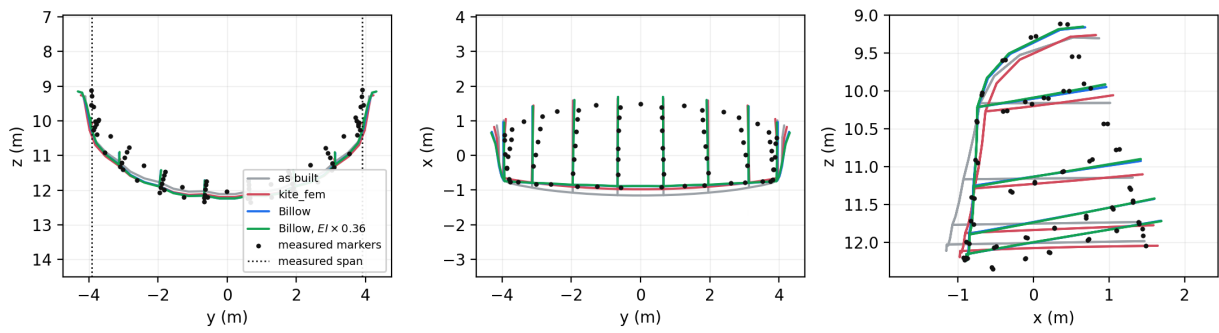


Figure 13: Load case 9.

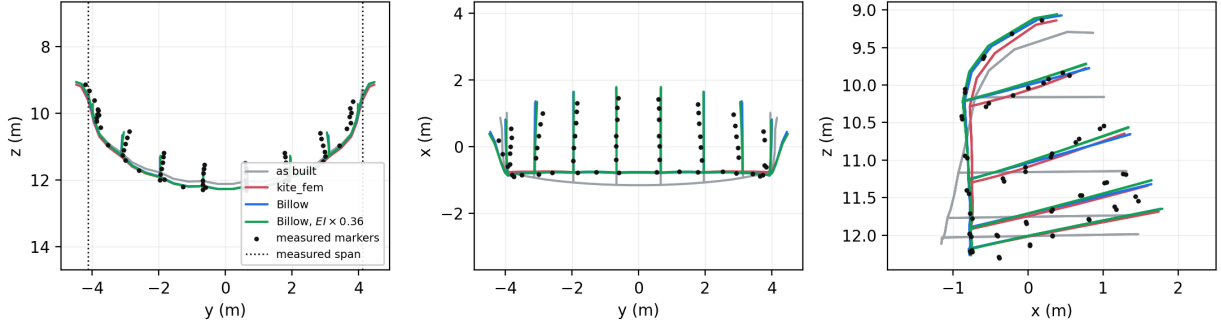


Figure 14: Load case 10.

4.3 The centre-load cases, which neither code reproduces

Cases 2, 3, 7 and 8 put 9.7 or 25.2 kg on the mid-span leading edge — a node with *no bridle attached*, carrying up to twice the kite’s own weight. Measurement has the span slightly *widening* against bare gravity (5.04 against 4.87 m): the kite tilts forward and the tips fold out.

Neither code reproduces that. `kite_fem` overstates the span by 34–48%, and its author is explicit about the cause: “by applying such a local load directly on the inflatable beam, the cross-section deforms and the stiffness of the tube reduces ... the simplification made by modelling such a complex structure as a beam element cannot capture this” (thesis §7.1). Billow fails differently and more visibly: it converges — to residuals of order 1×10^{-8} N — onto a *folded* branch, closing the span to under 1 m. Seeding from the gravity equilibrium and ramping the load in does not recover the physical branch. The energy is non-convex, and this is a second local minimum, not a numerical failure: a converged wrong answer.

These four cases are marked † in Table 1 and reported separately rather than averaged in, because a mean over them measures how each code fails rather than how well it works.

5 What this case actually tests

The comparison was built to isolate the canopy formulation, since that is the one element type that differs between the two codes. It does not do that. A sensitivity sweep on case 1:

change	span (m)
baseline	7.19
canopy Et 5000→1000 N/m	7.15
canopy deleted entirely ($Et = 1$ N/m)	7.36
tube moment–curvature curve scaled by 0.3	3.73

Deleting the canopy outright moves the answer less than a quarter as far as a modest change in tube bending. **This is a tube-bending test**, and any canopy conclusion read off it would be unsupported.

Collapse. None of the three codes models post-collapse behaviour. Breukels’ source data characterises it — the measured moment falls past collapse — but `kite_fem` computes a collapse flag and never acts on it (a literal `TODO: verify collapse` in `BeamElement.py`), and Billow saturates at $M_{\max}$ instead of dropping. A law that does follow collapse, fixed uniquely by requiring it to match both the fitted initial stiffness and the fitted collapse curvature,

$$W(\kappa) = EI_0 \kappa_c^2 \left[1 - \left(1 + \frac{\kappa}{\kappa_c} \right) e^{-\kappa/\kappa_c} \right], \quad M(\kappa) = EI_0 \kappa e^{1-\kappa/\kappa_c}, \quad (1)$$

converges without difficulty (56–77 iterations, ~ 3 s, residual 3×10^{-9}) and changes the answer by 20 mm. It changes nothing here because Billow’s beams reach only 31% of the collapse curvature and none of the 98 is flagged collapsed. Note also that collapse arrives at $0.7 \kappa_0$, i.e. at only $\sim 52\%$ of $M_{\max}$, so the saturating plateau is already too generous well before the cliff. It matters in general; it does not matter in this dataset.

6 Cost, convergence, and what the `kite_fem` numbers are

Table 4: Solver cost. * marks cases that never reached `kite_fem`’s own 0.01 tolerance. **The two time columns were measured on different machines** and are not a controlled benchmark — see the caveats below.

LC	kite_fem		Billow		
	t (s)	final tol.	t (s)	iterations	$\ r\ $ (N)
1	546	0.010	7	89	2e-07
2	671	0.010	15	200	9e-08
3	465	0.229*	19	228	4e-07
4	443	0.010	16	212	1e-07
5	619	0.012*	18	240	4e-07
6	576	0.010	6	86	4e-09
7	621	0.010	15	209	4e-09
8	468	0.076*	17	235	6e-09
9	483	0.010	15	187	1e-07
10	626	0.011	16	227	4e-07
mean	552		14	191	

Billow reaches force balance to better than 5×10^{-7} N on every case, against a target of 1×10^{-4} N, in a mean of 14 s. `kite_fem`’s recorded times average 552 s and three of its ten cases stop above its own tolerance (0.229, 0.076, and a marginal 0.0117).

Robustness required one addition to the solver. On slack-dominated configurations a near-zero-curvature search direction is answered with a step of order 1×10^4 , the objective overflows, and the line search collapses before restoration can help. A move limit — a trust region expressed as bounds IPOPT already enforces — cures this and reproduces the direct solve to the digit. One caveat came with it: once bounds are active, IPOPT reports success for the *boxed* problem while the iterate sits on the boundary, so the acceptance test must key off the force residual and never off the solver’s own verdict.

6.1 Caveats on every `kite_fem` number in this report

These matter enough to state plainly, because the comparison is the weakest part of this work and the reader should discount it accordingly.

It was never reproduced. Every `kite_fem` figure here comes from its *committed* `model_results.csv` and saved `.npz` states, not from a run performed for this study. One attempt was made; it produced no output in over ninety minutes and was abandoned. So nothing here distinguishes “`kite_fem` as published” from “`kite_fem` as it runs today”.

What *was* verified is narrower but load-bearing: our length extractor reproduces its published table from its own saved states to 1×10^{-9} m on all ten cases (a committed test). The rescoring is therefore sound even though the solve is not reproduced.

Its solver settings are tuning choices we did not replicate. The published run uses an `I_stiffness` of 15, a tolerance of 0.01, a step limit, a relaxation schedule, and an outer loop that calls `adapt_stiffnesses` — progressively stiffening any element above 1% strain, up to four times. None of that was re-derived or assessed here. A different setting could move its answers by an unknown amount.

It is not solving the model it states. Its converged beams sit at about $0.23\times$ its own constitutive law, with negative bending stiffness in three of the four cases inspected, and its anchor node belongs to no element so its equilibrium is held by `I_stiffness` rather than by a load path — its own solutions drift 0.29–0.34 m bodily. Comparing against it is therefore comparing against a numerical state, not a clean reference solution.

“`kite_fem`” is not one thing. The results used here are from `main` at 630c8c0. AWETrim’s own environment installs the branch `mass_proportional_regularisation` at 6edff17 — named for the very regularisation implicated above. Any statement in this report about `kite_fem` refers to the former.

The bridle is not matched in the headline comparison. `kite_fem` ran the bridle exactly as stored. Our primary results apply the Table B.1 corrections. Scored on the stored bridle for a like-for-like comparison, Billow gives 15.1% against `kite_fem`’s 12.4%; on the corrected bridle it gives 15.7%. The difference the bridle makes overall is therefore 0.6 percentage points — small, but the comparison is not bridle-matched and should not be read as though it were.

The metric is ours, not theirs. Its published `shapecorrelation` is not used, for the reason given in Table 1’s caption. Numbers here will therefore not match those reported upstream.

The timings are not a benchmark. Its recorded times are from an HP Zbook Power G7; ours are from a different machine, running Python against a compiled CasADi/IPOPT stack while `kite_fem` runs Cython `pyfe3d` elements. Read the ratio as an order of magnitude at most, and note that the two codes are not even solving to the same tolerance — a factor of 1×10^3 separates the residuals.

7 What stiffness the measurement implies

Rather than fit bridle lengths, the whole tube moment–curvature curve was scaled by a single factor λ — “`moment_max`”, “`bending_stiffness`” and “`torsion_c1`” together, so $\kappa_0 = M_{\max}/EI_0$ is unchanged and the curve keeps its shape while only its magnitude moves. The two bare-gravity cases, the only two that genuinely load the tubes, then say what stiffness the measurement wants:

λ	case 1, 0.15 bar		case 6, 0.25 bar	
	span (m)	error (%)	span (m)	error (%)
1.00	7.276	25.9	7.468	19.5
0.55	6.156	14.6	6.589	11.5
0.45	5.569	9.3	6.120	7.7
0.35	4.674	8.3	5.376	5.5
0.25	3.376	18.6	4.141	14.5
measured	4.867		5.453	

Both cases imply $\lambda \approx 0.36$, and — this is the part that matters — they imply the *same* factor at both pressures. A pressure-dependent error in the fit would not do that; a constant offset would. It is also further evidence against the low-pressure extrapolation of $M_{\max}$ discussed above, which would have required more softening at 0.15 bar than at 0.25 bar.

Run across all ten cases, a single $\lambda = 0.36$ gives a mean relative error of 8.0% against `kite_fem`’s 12.4%, and 6.2% against 11.5% on the two cases that test the tubes. It also repairs the centre-load cases, which the unscaled model got worst: case 2 falls from 27.6% to 9.6%, case 7 from 19.7% to 7.5%. The tip-load cases barely move (6.5%→6.6%, 5.2%→5.2%), exactly as the previous section predicts — they are bridle-determined and carry no information about the tube law.

This is an identification, not a fit. One scalar, applied uniformly to all 98 tube elements, calibrated on two cases and improving eight. It is reported because a consistent, pressure-independent factor is evidence about the fit rather than a free parameter absorbing error — but it is not adopted as a model change. Two things must be understood first.

Where does 0.36 come from? Inverting the fitted EI_0 through $EI = Et\pi r^3$ gives an implied Et of 133–155 kN/m for the leading-edge tubes, nearly constant across $d = 0.113 - 0.202$ m — so the fit’s radius dependence is physically consistent, and the implied fabric (0.5–1.2 GPa at 0.15–0.25 mm) is plausible to soft for polyester. The fit is therefore not obviously twice too stiff on material grounds, which makes a calibration or usage mismatch the more likely explanation.

Two candidates are worth checking against Breukels’ thesis directly. First, his tubes are discretised as *rigid cylindrical segments joined by torque springs*, and the C_1 – C_8 coefficients were fitted so that such a chain reproduces the measured beam — a discretisation-dependent calibration, not an intrinsic moment–curvature law, and re-using it as a continuum law could carry a systematic factor. Second, his nomenclature defines an *effective* internal pressure p_{eff} distinct from the gauge pressure; if the fits are calibrated on p_{eff} , feeding 0.15 bar gauge straight in is wrong at exactly this kind of magnitude.

8 How weakly the published data constrains a model

The twelve published quantities are less information than they look. Every left/right pair is identical to 0.0×10^0 — $\text{La} \equiv \text{Li}$, $\text{LcsTL} \equiv \text{LcsTR}$ — so the data is exactly mirrored and only six numbers are independent. The thesis itself scores just three: the trailing-edge length (the sum of $\text{La} - \text{Li}$), the span, and the tip-to-centre distance.

More limiting is what is absent. There is no height, no sag depth, and no chordwise measure of any kind: every published quantity lies in the spanwise / trailing-edge geometry. A model can match all three scored lengths and still have the wrong shape — the wrong camber, the wrong chordwise twist, the wrong depth of sag — with nothing in the dataset able to detect it. That is a real ceiling on what any validation against this file can claim, including this one.

The information exists. The measurement was stereoscopic, giving full three-dimensional marker positions, and the thesis plots measured geometry against model geometry in top, side and front projection for all ten load cases (its Figure 7.2). Only the twelve scalars were published.

The single most valuable thing to obtain is therefore the 3D marker coordinates behind Figure 7.2. They would turn this from a comparison of three effective scalars into a comparison of shapes, and would immediately settle questions the present data cannot touch — among them whether Billow’s wing is uniformly too open or wrong in a particular region, and whether the centre-load cases fail in the way the folded branch suggests.

9 The proper test: 3D shape against the markers

Everything above is scored on twelve numbers that reduce to six independent ones. Haanen’s marker clouds — 62 to 76 three-dimensional points per load case on the leading edge, the eight struts and the canopy — allow the question that data cannot answer: is the *shape* right?

Method. Markers are in the stereo rig’s frame, so a *rigid* transform is needed — rotation and translation only, never scale, since scale is what is being tested. Correspondence comes from structure rather than fitting: the measured struts are ordered across the span with their first marker at the leading edge, so their roots pair with the model’s eight strut leading-edge nodes, giving eight anchors for a Kabsch fit. Both spanwise orderings are tried, because the kite is near-symmetric and the marker numbering direction is undocumented. The fit is then refined by ICP against the model curves, so no marker is assumed to sit on a node. Reported errors are point-to-*surface*: the markers are stuck to the outside of the tubes while the model polyline is the tube axis, and ignoring that radius — 55 to 100 mm here — would put a systematic offset into every case.

The scaled model reaches the measurement floor. On the two cases that load the tubes, the error falls from 480 to 100 mm and from 387 to 101 mm — onto the same ~ 100 mm the tip cases sit at, which is the marker placement and alignment floor. Across all ten the mean falls from 315 to 125 mm, against 404 mm for doing nothing. That the factor identified from two scalar spans also fixes the full three-dimensional shape, including the centre-load cases it was not fitted on, is the strongest evidence in this report that it is a real property of the tube law rather than a number absorbing error.

The tip-load cases are confirmed uninformative. Their errors are 141, 102, 114, 101 mm for the *undeformed* geometry against 144, 101, 119, 102 mm for the unscaled model and 136, 92, 111, 94 mm for the scaled one — all within about 10 mm of each other. Doing nothing scores as well as either model. This is the same conclusion the bridle-tension argument reached in Section 8, now established directly in three dimensions rather than inferred, and it is why the six-case and ten-case means elsewhere in this report should be read with care.

Table 5: Point-to-surface RMS between the solved shape and the photogrammetry markers. “As built” is the undeformed geometry, included as the null model: anything not beating it has explained nothing.

LC	load	Billow (mm)		as built (mm)
		EI x1.0	EI x0.36	
1	gravity	480	100	695
2	centre 9.7 kg	501	152	654
3	centre 25.2 kg	507	171	639
4	tip 2 kg	144	136	141
5	tip 5 kg	101	92	102
6	gravity	387	101	554
7	centre 9.7 kg	399	151	525
8	centre 25.2 kg	409	145	516
9	tip 2 kg	119	111	114
10	tip 5 kg	102	94	101
mean, gravity only		434	101	625
mean, centre load		454	155	583
mean, tip load		117	108	115
mean, all ten		315	125	404

10 Summary

- The distributed reference model is ill posed — its anchor belongs to no element — and its published results are held up by a numerical regularisation. Any future use of this dataset has to supply a suspension.
- `kite_fem`’s beams converge to $\sim 0.23\times$ its own constitutive law, with negative bending stiffness in several cases. Its closer agreement with the measurement is not evidence of better physics.
- The dataset is a tube-bending test — and only two of its ten cases even are. It cannot decide a canopy model, which was the original reason for building the comparison. The 3D markers confirm this directly: on the tip-load cases the undeformed geometry scores as well as either model.
- Against the 3D markers, $\lambda = 0.36$ takes the mean point-to-surface error from 315 to 125 mm (null model 404 mm), and brings the two tube-loaded cases onto the ~ 100 mm measurement floor — having been identified on two scalar spans, not on the shape.
- Post-collapse behaviour is unmodelled in all three codes and does not matter here or in coupled use, because the tubes never leave their initial slope.
- The residual disagreement lives in EI_0 . A single pressure-independent factor $\lambda \approx 0.36$, identified on the two gravity cases, improves eight of the ten and beats `kite_fem` overall (8.0% against 12.4%). It is reported as an identification, not adopted: two specific mismatches in how the Breukels fit is being re-used — its torque-spring discretisation and its effective pressure — should be checked against the source first.

Data and code

Measurements, the reference model and `kite_fem`’s published results are vendored under `data/LEI-V3-KITE/hang` from `awegroup/kite_fem` (MIT, © 2025 Patrick Roeleveld), `main` at 630c8c0. The model builder

is `scripts/structural/hanging_kite.py`; the driver, plots and this report's tables are the neighbouring `run_`, `plot_` and `make_hanging_report.py` scripts.

Thesis references are to P.I.H. Roeleveld, *Fast finite element modelling of bridled leading-edge inflatable kites*, MSc thesis, TU Delft, February 2026.